

**B.Sc in Computer Science and Engineering Thesis
Report on**

**Poly-Dialectal Neural Machine
Translation System for Bangla
Regional Dialects**

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22 August, 2026

1 Introduction

1.1 Background and Motivation

Bangla (Bengali), an Indo-Aryan language, stands as one of the most widely spoken languages globally, boasting over 240 million native speakers primarily distributed across Bangladesh and the Indian states of West Bengal, Tripura, and Assam. Despite its immense socio-linguistic prominence and rich literary heritage, Bangla is not a monolithic entity; rather, it exhibits substantial intra-language diversity through a vast array of distinct regional dialects. Dialects such as Sylheti, Chittagonian, Barisali, Noakhali, Rangpuri, Rajshahi, and Mymensingh differ markedly from Standard Colloquial Bangla (SCB) across multiple linguistic strata. These divergences manifest as profound phonological variations, complex morphological inflections, altered syntactic structures, and highly localized lexical choices. In extreme cases, such as the Chittagonian or Sylheti dialects, the linguistic deviation is so severe that they are often bordering on mutual unintelligibility with Standard Bangla.

This deep intra-language divergence engenders significant communication barriers within the broader socio-economic framework. Native speakers of peripheral regional dialects frequently encounter difficulties in seamless communication with speakers of other dialects, as well as in formal interactions, legal proceedings, and educational instructions traditionally conducted in Standard Bangla. Furthermore, modern Natural Language Processing (NLP) tools, machine translation systems, and digital interfaces—ranging from virtual assistants to automated governance platforms—are overwhelmingly optimized exclusively for Standard Bangla. Consequently, millions of dialectal speakers face systemic digital marginalization and language-access inequality, lacking automated communication technologies tuned to their native spoken forms. This technological disparity highlights an urgent need for inclusive NLP systems that can bridge the communication gap without enforcing linguistic homogenization.

1.2 Problem Statement

In recent years, Neural Machine Translation (NMT) architectures and Large Language Models (LLMs) have achieved state-of-the-art results for cross-lingual tasks involving high-resource languages. However, their efficacy degrades severely when applied to low-resource intra-language dialectal shifts. Most existing Bangla NLP benchmarks and pre-trained models treat the language as a singular, uniform distribution, fundamentally ignoring the complex cross-dialectal matrix.

This oversight stems from a confluence of critical technical challenges. Firstly, there is an acute scarcity of multi-dialectal parallel corpora required to effectively fine-tune deep neural networks. Secondly, the lack of standardized orthography for regional dialects exacerbates tokenization mismatches; standard Byte-Pair Encoding (BPE) [17] or Word-Piece [24] algorithms optimized for SCB frequently fragment dialectal words into meaningless subwords, destroying semantic integrity. Finally, existing regional text datasets suffer from severe class imbalance and are predominantly tailored for simple classification tasks rather than complex generative alignment. Addressing these challenges requires a paradigm shift from uni-dialectal modeling to a holistic, poly-dialectal approach.

1.3 Proposed System and Contributions

To address these fundamental limitations, this research introduces a unified **Poly-Dialectal Neural Machine Translation System** tailored specifically for the morphological and syntactic nuances of Bangla regional dialects. As illustrated in Figure 1, our architecture enables robust, multi-directional translation across three core operational paradigms: Dialect $\rightarrow$ Standard Bangla, Standard Bangla $\rightarrow$ Dialect, and direct Dialect $\rightarrow$ Dialect translation, thereby eliminating the cascading errors typically introduced by pivot-based translation methods.

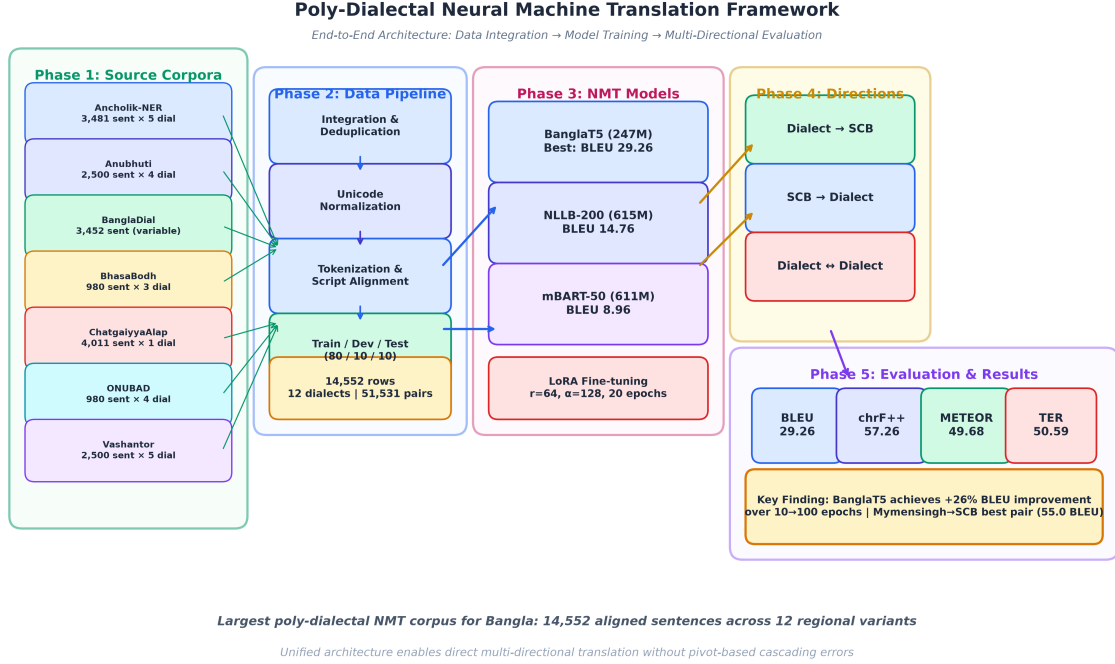


Figure 1: End-to-end architectural workflow of the Poly-Dialectal Neural Machine Translation Framework, depicting five phases: source corpora integration from seven existing datasets, the data processing pipeline, NMT model fine-tuning with LoRA, multi-directional translation paradigm, and evaluation with four standard MT metrics.

The primary contributions of this thesis are summarized as follows:

1. **Largest Multi-Dialectal Parallel Corpus:** We construct the most expansive multi-dialect machine translation corpus for Bangla to date, comprising 14,552 aligned rows across 12 regional dialect columns. The corpus integrates seven prior datasets—Ancholik-NER, Anubhuti, BanglaDial, BhasaBodh, ChatgaiyyaAlap, ONUBAD, and Vashantor. Additionally, we incorporate 500 manually created, bidirectional, and verified high-quality parallel sentence pairs for five dialects (Rangpur, Tangail, Kishoreganj, Narail, Narsingdi), yielding over 51,531 non-null parallel pairs across 11 regional variants.
2. **Pioneering Benchmark of NMT Architectures on Poly-Dialectal Translation:** To the best of our knowledge, this work presents the first systematic empirical evaluation of prominent sequence-to-sequence architectures—specifically BanglaT5 [2], NLLB-200 [22], and mBART-50 [21]—on the novel task of multi-directional, poly-dialectal Bangla translation.

3. **Systematic Cross-Dialectal Linguistic Analysis & Scaling Study:** We conduct a rigorous Optimal Dataset Size Study (progressively scaling from 5,000 instances) and quantitative characterization of cross-dialectal transfer mechanisms. We uncover how morphological proximity and training data volume jointly influence neural machine translation efficacy across diverse metrics.
4. **Real-World Web Deployment:** We deploy the best-performing NMT model in a fully functional web application, bridging the digital language divide by offering real-time poly-dialectal translation capabilities accessible to end users.

1.4 Thesis Organization

The remainder of this report is organized as follows: Section 2 critically reviews related literature on Bangla regional NLP and dialectal machine translation. Section 3 formally states the research hypothesis and objectives. Section 4 details the methodology, including dataset construction, alignment strategies, and model configurations. Section 5 presents a comprehensive evaluation of the experimental results. Section 6 discusses the limitations of the current framework and proposes future research directions, followed by concluding remarks in Section 7.

2 Literature Review

2.1 Overview of Regional Bangla NLP

Historically, the development of linguistic resources for Bangla has heavily favored Standard Colloquial Bangla, leaving regional dialects severely underrepresented in the NLP ecosystem. However, over the past few years, interest in low-resource regional Bangla NLP has seen a significant upsurge. Initial studies focused predominantly on resource creation for text classification and sentiment analysis tasks for specific regions.

Sultana et al. [20] established the *BdRegionText* corpus, a pioneering effort for regional text classification, which starkly demonstrated the performance bottleneck of standard classifiers when exposed to rich dialectal vocabulary. Building on the need for broader representation, Mahi et al. [11] released *BanglaDial*, a merged text dataset that brought to light the severe class imbalance issues and data scarcity prevalent across different regional dialects.

In specialized downstream NLP tasks, the complexity of dialectal morphology continues to pose significant hurdles. Kundu et al. [9] introduced *ANUBHUTI*, a dedicated corpus targeting sentiment analysis in regional languages, revealing that emotion markers vary significantly across regional boundaries. For the crucial task of offensive speech identification, Fayaz et al. [6] developed *BIDWESH*, evaluating hate speech detection mechanisms across regional dialects and proving that SCB-trained toxicity classifiers fail to generalize to dialectal slang. Concurrently, Paul et al. [14] formulated *ANCHOLIK-NER*, providing essential named entity annotations across regional variants to facilitate information extraction in non-standard texts.

2.2 Dialectal Machine Translation and LLM Benchmarks

While text classification laid the groundwork, direct efforts in generative machine translation for Bangla dialects have emerged only recently, driven by the advent of massively multilingual neural models. Early translation corpora were highly localized. For instance, Chowdhury et al. [4] built *ChatgaiyyaAlap*, a dedicated parallel corpus strictly designed for translating Chittagonian dialect sentences to Standard Bangla. To expand the dialectal scope, Sultana et al. [19] presented *ONUBAD*, providing an initial generalized resource for automated regional dialect conversion. This was followed by Faria et al. [5], who introduced *Vashantor*, a multilingual benchmark covering five regional dialects. Though a significant step forward, *Vashantor* remained constrained by its limited volume, offering only 2,500 sentences per dialect.

More recently, the research community has pivoted towards exploring the zero-shot and few-shot capabilities of Large Language Models (LLMs) for dialect translation. Mahjabin et al. [12] constructed *BanglaCHQ-Prantik* to benchmark state-of-the-art LLMs against human-evaluated dialect translation standards. Similarly, Jawad et al. [8] presented *DIALTSA-BN*, evaluating LLM capabilities in joint dialect translation and sentiment analysis. Furthermore, Bhuiyan et al. [3] explored *BhasaBodh*, investigating methods to bridge Bangla dialects and romanized forms through advanced machine translation techniques. Despite these advancements, fine-tuned NMT models consistently outperform generalized LLMs in specialized low-resource dialectal translation due to the LLMs' inherent lack of dialect-specific pre-training data.

2.3 Critical Synthesis and Research Gap

While the aforementioned contributions have collectively laid a valuable foundation for Bangla dialectal NLP, a critical review reveals several major structural and methodological shortcomings that currently stifle the development of robust, production-ready translation systems:

1. **Single-Dialect Isolation:** Corpora such as *ChatgaiyyaAlap* focus exclusively on a single dialect pair (e.g., Chittagonian-Standard). Models trained on such isolated datasets fail entirely to generalize across other regional dialects, necessitating the training of separate, disjoint models for every region.
2. **Severe Corpus Volume Limitations & Lack of Scaling Studies:** Multi-dialect datasets like *Vashantor* offer only modest data volumes (e.g., 2,500 samples per dialect). This scale is vastly insufficient for deep fine-tuning of architectures like mT5 or mBART-50. Furthermore, prior literature entirely lacks systematic scaling studies to empirically determine optimal dataset size thresholds for cross-dialectal transfer.
3. **The Pivot Translation Problem:** Existing works lack a unified framework capable of translating directly between different regional dialects (e.g., Sylheti $\rightarrow$ Chittagonian). Current systems force a “pivot” approach, routing translation through Standard Bangla as an intermediary, which compounds translation errors and degrades final semantic accuracy.
4. **Absence of Real-World Deployment:** State-of-the-art models developed in prior studies remain confined to offline evaluation environments. The community lacks open, deployed web applications that can provide real-time translation services directly to marginalized dialect speakers.

To definitively resolve these limitations, our work introduces a comprehensive, balanced, poly-dialectal parallel corpus that is orders of magnitude larger than previous efforts. By fine-tuning state-of-the-art Transformer models on this expansive dataset, we enable direct, multi-directional, poly-dialectal translation within a single, unified neural architecture.

2.4 Summary of Dialectal Resources

Table 1 summarizes the major dialectal datasets and resources reviewed in this section, detailing their coverage, architecture, evaluation metrics, and state-of-the-art performance.

Table 1: Comprehensive Summary of Bangla Regional Dialect NLP Resources and Datasets

Dataset & Year	Primary Task & Domain	Dialects & Dataset Scale	Methodology & Quality Control	Key Metrics & SOTA Findings
1. ANCHOLIK-NER (2025)	Dialectal Named Entity Recognition (NER); 10 entity classes.	Sylhet, Chittagong, Barishal. 10,443 sentences (3,481/dialect).	BIO tagging. Sourced from Vashantor (71.8%), ONUBAD, & news. 3-annotator consensus pipeline.	First dialectal NER baseline. Assamese NER baseline reached 80.69% F1 (MuRIL); medical BanglishBERT got 56.13% F1.
2. ANUB-HUTI (2026)	Thematic Sentiment & Multilabel Emotion Analysis in dialectal text.	Mymensingh, Noakhali, Sylhet, Chittagong. 10,000 sentences (2,500/region).	Sourced from MONOVAB. 8 native translators (2/region) with 4-stage QA (sentiment, NaN parsing, anomaly, spelling).	High inter-rater agreement (Cohen’s $\kappa = 0.76\text{--}0.84$). Dominant emotions: Contempt (921), Disgust (871), Anger (732).
3. BIDWESH (2025)	Multi-Dialectal Hate Speech & Toxicity Detection.	Barishal, Noakhali, Chittagong. 9,183 instances (3,061 parallel/dialect).	Sourced from BD-SHS; translated by 6 native speakers; 5-stage validation. Target/Type taxonomy.	Balanced binary split (49.2% Hate / 50.8% Non-Hate). Individual targets highest (34.4%); Slander dominant (53.7%).
4. Bangla Dialect Bibliography	Systematic Bibliography & Meta-Review of Bangla Dialect NLP.	Comprehensive overview across major regional variants in BD.	Systematic synthesis tracking 11 publications, annotation strategies, & baselines.	Mapped 11 unique corpora; highlights Sylheti & Chittagonian as most studied; Barishal, Mymensingh, Noakhali emerging.
5. BanglaDial (2025)	Dialect Identification (DID) & Class Imbalance Modeling.	11 Dialects + Standard Bangla. 60,729 sentences (362,430 words).	Aggregated from 4 corpora (Vashantor, ONUBAD, etc.). Deduplicated (4.1% removed) & normalized.	Long-tail distribution (Top-3: 41.3% of records). Imbalance Ratio $IR = 9.82$; Shannon Entropy $H = 3.416$ bits.

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Dataset & Year	Primary Task & Domain	Dialects & Dataset Scale	Methodology & Quality Control	Key Metrics & SOTA Findings
6. BdRe-gionText (2022)	Bangla Regional Text Classification on social media content.	Chittagong, Noakhali, Barishal, Rangpur. 2,573 sample texts.	Scraped from FB/YouTube/IG with human validation. Extracted TF-IDF, CountVectorizer, & BoW.	Random Forest + TF-IDF achieved peak 79.15% accuracy (10-fold CV: 79.47%, weighted F1: 0.78). RF + BoW got 75.92%.
7. BhasaBodh (2025)	Script Normalization & Multilingual Dialect Translation (Native/Romanized).	Chittagong, Sylhet (aligned with Standard Bangla & English). 1,960 sentences.	Sourced from ONUBAD; augmented via Gemini 2.5 Pro for Romanizations. Fine-tuned NLLB-200 & mBART-50.	mBART-50 achieved peak 87.44 BLEU on Romanized → Standard Bangla and 74.36 BLEU on Chittagong ↔ Sylhet.
8. Chat-gaiyyaAlap (2025)	Dialect Conversion (Chittagonian → Standard Bangla).	Chittagonian (aligned with Standard Bangla). 4,012 sentences + 1,500-word dictionary.	Sourced from social media/dramas; validated by 5 native Chittagonian graduates.	Created clean translation mapping resolving severe grammatical divergences & phonetic shifts in negative sentences.
9. BanglaCHQ-Prantik (2025)	Medical-Domain Machine Translation (Standard Bangla → Dialects).	Sylheti (2,350 sentences), Chittagonian (500 sentences).	Sourced from BanglaCHQ-Summ; validated by 11 validators & 2 physicians. Evaluated LLMs (Qwen, Gemma, GPT-4o, Gemini).	Gemini 2.5 Flash achieved SOTA on Sylheti (23.67 BLEU); GPT-4o with CoT achieved SOTA on Chittagonian (24.67 BLEU).
10. DIALTSA-BN (2025)	Dialect Translation & Sentiment Analysis (Native vs. Transliterated).	Chattogram, Barishal, Sylhet, Noakhali. 600 annotated instances (150/region).	Sourced from YouTube; filtered via Word2Vec. Evaluated Claude 3.7, GPT-4o-mini, Gemma-3, Llama-3.1, etc.	Transliteration boosted translation BLEU from 0.07 to 0.330 (Claude 3.7) & 0.317 (GPT-4o-mini). Few-shot sentiment F1 reached 0.98.

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Dataset & Year	Primary Task & Domain	Dialects & Dataset Scale	Methodology & Quality Control	Key Metrics & SOTA Findings
11. BanglaCHQ-Prantik Baseline (2025)	Pediatric & Adult Clinical MT Error Taxonomy.	Sylheti, Chittagonian. 2,850 total aligned queries.	Parallel clinical reference tracking. Highlighted phonetic shifts (e.g., *pete* → *feto* / *cano*).	Identified 4 main LLM failure modes: terminology mismatch, dosage corruption ("500mg" → "5 gram"), idioms, & orthographic drift.
12. ONUBAD (2025)	Fine-Grained Dialect Conversion & Neural Machine Translation.	Chittagong, Sylhet, Barisal. 7,950 total parallel rows (980 sen- tences/region).	Sourced from liter- ature/volunteers; validated by university linguists. Benchmarked Seq2Seq & Transformers.	Documented 2 words/phrase & 3 words/sentence. BanglaT5 and SeamlessM4T established peak NMT baselines.
13. Vashantor (2023)	Multi-Task Dialectal NMT & Region Clas- sification.	Chittagong, Noakhali, Sylhet, Barishal, Mymensingh. 32,500 total sentences.	Sourced from social media/news; validated with Cohen's & Fleiss' Kappa. Custom DialectBanglaT5 & Dialect- BanglaBERT.	DialectBanglaT5 achieved SOTA on Mymensingh (71.93 BLEU). DialectBanglaBERT reached 89.02% region classification accuracy.

3 Research Hypothesis and Objectives

3.1 Research Hypothesis

We posit the following hypothesis: *A neural machine translation model trained on a unified, multi-dialectal parallel corpus encompassing multiple Bangla regional variants will demonstrate significantly improved generalization and cross-dialectal transfer compared to models trained on isolated, uni-dialectal datasets.* Specifically, we hypothesize that a poly-dialectal training paradigm exploits shared morphological and syntactic structures across related dialects, enabling the model to learn transferable representations that benefit even low-resource dialect pairs.

3.2 Research Objectives

The specific objectives of this research are:

1. To construct the largest multi-dialect parallel corpus for Bangla machine translation by integrating, aligning, and deduplicating seven existing dialectal datasets across 12 regional variants.
2. To develop a unified poly-dialectal translation system capable of three translation paradigms: Dialect $\rightarrow$ Standard Bangla, Standard Bangla $\rightarrow$ Dialect, and direct Dialect $\rightarrow$ Dialect translation.
3. To systematically evaluate and compare the performance of three state-of-the-art multilingual NMT architectures—BanglaT5 [2], NLLB-200 [22], and mBART-50 [21]—under parameter-efficient LoRA [7] fine-tuning on the constructed corpus.
4. To quantitatively analyze cross-dialectal transfer effects by examining how training data volume and linguistic proximity jointly influence translation quality.
5. To promote digital inclusion and linguistic accessibility for Bangla dialect speakers by establishing open benchmarks for future research.

4 Methodology

4.1 Dataset Construction and Integration

The foundation of our poly-dialectal NMT system is a meticulously curated multi-dialect parallel corpus. We integrate seven publicly available dialectal datasets, each contributing varying levels of coverage across different regional variants:

Table 2: Source corpus contributions per dialect. Values indicate the number of parallel sentence pairs available from each source.

Source Corpus	SCB	Syl	Cht	Bar	Mym	Noa	Ran	Raj
Ancholik-NER	3,481	3,481	3,481	3,481	3,481	3,481	0	0
Anubhuti	2,500	2,500	2,500	0	2,500	0	0	0
BanglaDial	3,452	442	577	790	712	0	655	891
BhasaBodh	980	980	980	0	0	0	0	0
ChatgaiyyaAlap	4,011	0	4,011	0	0	0	0	0
ONUBAD	980	980	980	980	0	0	0	0
Vashantor	2,500	2,500	2,500	0	2,500	2,500	0	0
Total	13,556	6,422	10,567	4,270	5,712	5,000	1,155	891

The final integrated dataset comprises 14,552 rows across 12 dialect columns (Standard Bangla plus 11 regional variants: Sylheti, Chittagonian, Barisali, Mymensingh, Noakhali, Rangpuri, Rajshahi, Kishoreganj, Narail, Narsingdi, and Tangail), yielding 51,531 total non-null parallel sentence pairs. This represents the largest such corpus for Bangla to date. Crucially, to mitigate the extreme scarcity of data for certain regional variants, we manually created a high-quality dataset consisting of altogether 2,500 bidirectional parallel sentence pairs (Standard Bangla $\leftrightarrow$ Dialect) distributed evenly across five dialects: Rangpur, Tangail, Kishoreganj, Narail, and Narsingdi (500 pairs each). To ensure rigorous data quality, each of these dialectal subsets was independently evaluated and verified by three native speakers of the corresponding dialect.

As depicted comprehensively in Figure 2, despite augmentation efforts, the dataset exhibits significant class imbalance—Chittagonian commands 10,567 entries while the newly added dialects (e.g., Narail, Tangail) are constrained to approximately 500 pairs. This unified visualization delineates both the gross dialectal volume and the exact decomposition of contributing source corpora for each variant. Furthermore, Figure 3 presents the cross-dialect parallel coverage matrix, quantitatively illustrating the pairwise intersection of translated sentences across all regional variants.

4.2 Data Preprocessing Pipeline

The raw integrated corpus undergoes a rigorous four-stage preprocessing pipeline:

1. **Unicode Normalization:** All text is normalized to Unicode NFC form to resolve inconsistent character compositions prevalent in Bangla script.
2. **Cleaning and Deduplication:** Duplicate entries across source datasets are identified and removed. Empty, whitespace-only, and non-Bangla script entries are filtered.

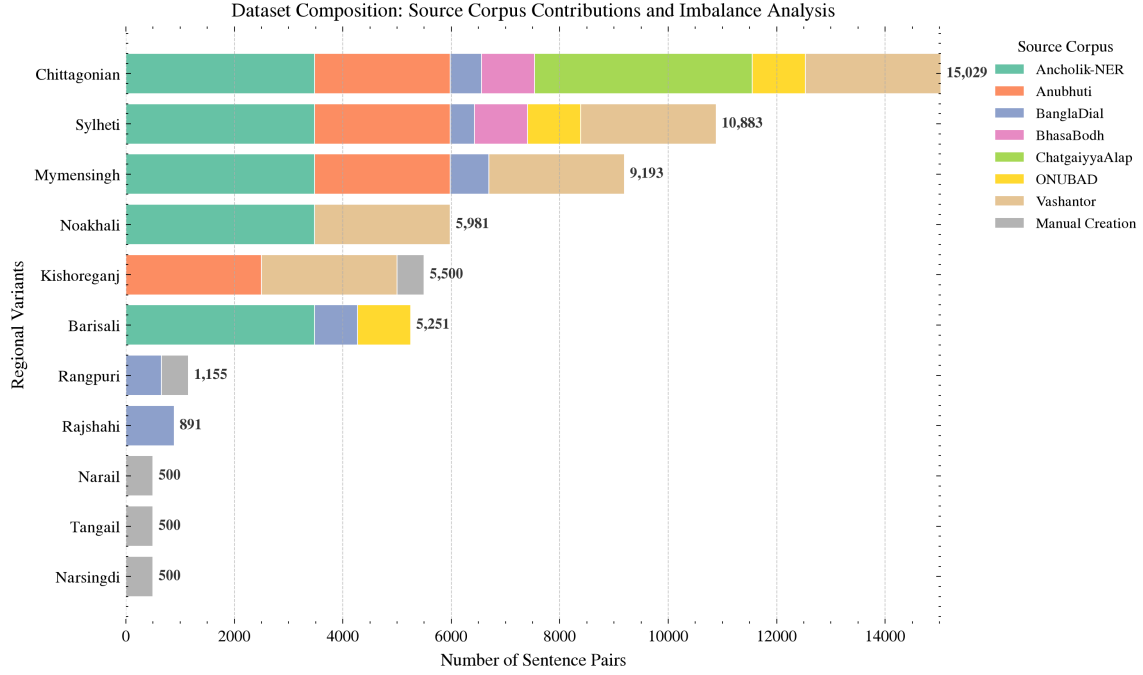


Figure 2: Dataset Composition and Source Provenance: A stacked horizontal distribution illustrating the severe class imbalance alongside the precise proportional contributions from prior corpora and manual augmentation efforts for each regional dialect.

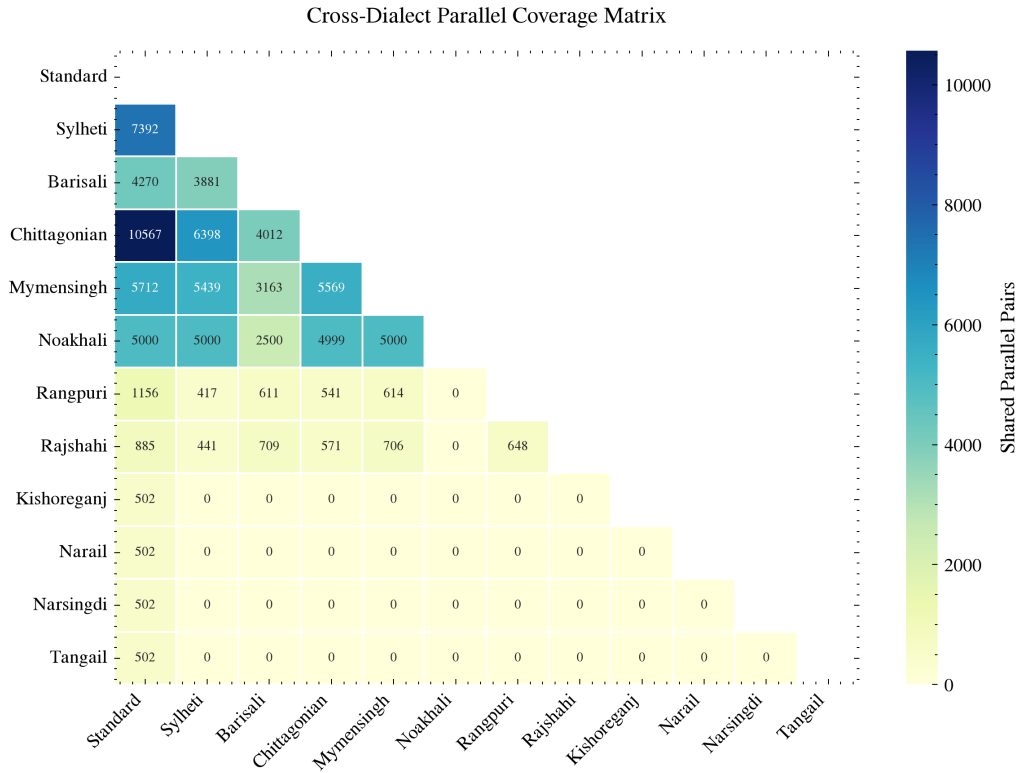


Figure 3: Cross-Dialect Parallel Coverage Matrix: An upper-triangular masked heatmap detailing the exact density of shared, parallel sentence pairs intersecting any two evaluated dialectal variants.

3. **Tokenization and Script Alignment:** Text is tokenized using model-specific SentencePiece tokenizers. For BanglaT5, the 32K vocabulary is optimized for Bangla NLG tasks; for NLLB-200 and mBART-50, the larger vocabularies ($\sim 256K$ and $\sim 250K$, respectively) accommodate multilingual coverage.
4. **Train/Dev/Test Split:** The corpus is partitioned into training (80%), development (10%), and test (10%) sets with stratified sampling to ensure proportional dialect representation across splits.

4.3 Model Architectures

We evaluate three state-of-the-art encoder-decoder Transformer [23] architectures, chosen to represent a spectrum of design philosophies for multilingual and Bangla-specific NMT. The architectural comparison is summarized in Table 3.

Table 3: Architectural comparison of the three NMT models evaluated in this study.

Feature	BanglaT5	NLLB-200	mBART-50
Parameters	247M	615M	611M
Architecture	T5	Transformer	BART
Encoder/Decoder Layers	12/12	12/12	12/12
Hidden Dimension	768	1,024	1,024
Attention Heads	12	16	16
FFN Dimension	3,072	4,096	4,096
Vocabulary Size	32K	$\sim 256K$	$\sim 250K$
Languages Supported	Bangla	200	50

4.3.1 BanglaT5

Overview and Capabilities: BanglaT5 [2] is a sequence-to-sequence model based on the T5 [16] architecture, pre-trained extensively on approximately 27.5 GB of monolingual Bangla text. **Advantages and Limitations:** Being a language-specific model, its 32K vocabulary is heavily optimized for Bangla morphology, preventing the excessive subword fragmentation commonly seen in multilingual tokenizers. However, it lacks cross-lingual pre-training signals that might otherwise aid in zero-shot generalization. **Suitability for Dialects:** We selected BanglaT5 because its deep grounding in Standard Bangla provides a strong prior for generating morphologically complex dialectal variations. It serves as the primary language-centric baseline in our study.

4.3.2 NLLB-200

Overview and Capabilities: NLLB-200 [22] (No Language Left Behind) is Meta AI’s massively multilingual Transformer designed for direct translation across 200 languages. We utilize the distilled 600M parameter variant. **Advantages and Limitations:** Its primary advantage is robust multilingual alignment and a vast $\sim 256K$ vocabulary designed to encompass diverse scripts. However, this large vocabulary dilutes its representational capacity for any single low-resource language, potentially hampering its nuance in intra-language dialectal shifts. **Suitability for Dialects:** NLLB-200 is selected to determine if

massive multilingual pre-training provides implicit transfer learning benefits for regional dialects.

4.3.3 mBART-50

Overview and Capabilities: mBART-50 [21] is a multilingual denoising auto-encoder based on the BART [10] architecture, extended to cover 50 languages. **Advantages and Limitations:** It excels at sequence generation tasks due to its denoising objective during pre-training. Like NLLB-200, its limitation lies in the generalized multilingual tokenization which is not exclusively tuned for Bangla scripts. **Suitability for Dialects:** We include mBART-50 as a strong baseline for multilingual generation to compare against both the language-specific BanglaT5 and the massively multilingual NLLB-200.

4.4 Hyperparameters

To ensure rigorous reproducibility and optimal convergence, Table 4 outlines the unified architectural and fine-tuning configurations applied across all three candidate models.

Table 4: Unified architectural and fine-tuning configurations for the NMT models.

Category	Hyperparameter	BanglaT5	mBART-50	NLLB-200
Model Registry	HuggingFace Checkpoint	csebuetnlp/banglat5_small	facebook/mbart-large-50-many-to-many-mmt	facebook/nllb-200-distilled-600M
	Base Parameter Count	~60M (Base) / 247M	~611M	~600M
	Src/Tgt Language Tags	Prefixd	bn_IN	ben_Beng
Sequence Setup	Max Length (L_{src} , L_{tgt})	128 tokens	128 tokens	128 tokens
	Sequence Packing	Enabled	Disabled	Disabled
Optimization	Optimizer	Paged AdamW 8-bit	Paged AdamW 8-bit	Paged AdamW 8-bit
	Learning Rate (η)	5×10^{-4}	5×10^{-4}	5×10^{-4}
	LR Scheduler	Cosine Annealing	Cosine Annealing	Cosine Annealing
Batching	Per-Device Batch Size	32	8	8
	Gradient Accum. Steps	2	8	8
	Effective Batch Size	64	64	64
PEFT / DoRA	Adapter Rank (r)	64	64	64
	Scaling Factor (α)	128	128	128
	Target Projections	q, v, k, o, wi, wo	q-proj, k-proj, v-proj, out-proj, fc1, fc2	q-proj, k-proj, v-proj, out-proj, fc1, fc2

4.5 Dataset Scaling Protocol

To formally investigate the relationship between data volume and translation generalization, we conduct an optimal dataset size study. We construct a fully parallel dataset encompassing Standard Bangla and four major dialects: Sylheti, Noakhali, Chittagong, and Mymensingh. The dataset is scaled iteratively from 500 to 4,499 parallel sentence pairs. The detailed hyperparameter matrix governing this scaling experiment is provided in Table 5.

Table 5: Hyperparameter matrix for dataset scaling experiments.

Parameter Class	Hyperparameter	Configuration
Model Config	Base Model	csebuetnlp/banglat5
Sequence Dims	Max Sequence Length	128 tokens (Source/Target)
PEFT Settings	DoRA Enabled	Yes ($r = 8, \alpha = 16$ & $r = 64, \alpha = 128$)
Optimization	Optimizer	Paged AdamW 8-bit
	Learning Rate (η)	5×10^{-4} (Cosine Decay)
Batching	Effective Batch Size	90 (Batch size 45, Grad. Accum. 2)
Training Schedule	Epochs per Step	10 Epochs

4.6 Fine-Tuning Strategy

All models are fine-tuned using **Low-Rank Adaptation (LoRA)** [7], a parameter-efficient method that injects trainable low-rank decomposition matrices into each attention layer while freezing the pre-trained weights. Our experimental pipeline consists of two distinct stages:

4.6.1 Stage 1: Best Model Selection

In the initial stage, all three candidate models (BanglaT5, NLLB-200, mBART-50) are trained for **20 epochs** using the complete integrated dataset. To ensure a fair comparative analysis, identical LoRA configurations (Configuration B: rank $r = 64$, $\alpha = 128$) are applied across all models. The best-performing model is then selected based on overall evaluation metrics for the next stage.

4.6.2 Stage 2: Final Training

In the second stage, the selected best model (BanglaT5) is retrained for an extended period of **100 epochs** to deeply analyze learning behavior and performance saturation. Due to hardware resource constraints (the use of a Kaggle platform featuring dual T4 GPUs and 32GB RAM), our compute resources were limited, enabling us to train only the top-performing model up to 100 epochs. This extended training phase facilitates detailed error analysis and linguistic pairwise comparisons to identify dialectal proximities.

4.7 Multi-Directional Translation Paradigm

Unlike prior uni-directional approaches, our system is trained to handle all possible translation directions within a single unified model:

- **Dialect $\rightarrow$ Standard Bangla:** Converting regional dialect text to SCB.
- **Standard Bangla $\rightarrow$ Dialect:** Generating dialectal text from SCB input.
- **Dialect $\rightarrow$ Dialect:** Direct translation between two different regional dialects without an intermediary pivot through Standard Bangla.

4.8 Evaluation Metrics

Model performance is assessed using four complementary automatic evaluation metrics. Given the morphological complexity of Bangla dialects, relying on a single metric is insufficient; thus, we utilize a combination of word-based, character-based, and edit-distance metrics.

4.8.1 BLEU (Bilingual Evaluation Understudy)

BLEU [13] measures the n -gram precision of the candidate translation against reference translations, applying a brevity penalty (BP) to discourage overly short outputs.

$$\text{BLEU} = \text{BP} \cdot \exp \left(\sum_{n=1}^N w_n \log p_n \right) \quad (1)$$

Interpretation and Appropriateness: Higher BLEU scores indicate greater n -gram overlap. While BLEU is the industry standard allowing for broad benchmarking, its limitation lies in its strict exact-word matching, which heavily penalizes legitimate morphological variations prevalent in Bangla dialects.

4.8.2 chrF++ (Character n -gram F-score)

chrF++ [15] computes the F -score based on character n -grams (typically up to 6-grams) combined with word-level bigrams.

$$\text{chrF++} = (1 + \beta^2) \frac{\text{chrP} \cdot \text{chrR}}{\beta^2 \cdot \text{chrP} + \text{chrR}} \quad (2)$$

where chrP and chrR represent character n -gram precision and recall, respectively. **Interpretation and Appropriateness:** A higher chrF++ score denotes better morphological alignment. This metric is exceptionally appropriate for highly inflected languages like Bangla, as it rewards partial word matches (e.g., matching root verbs with slightly altered dialectal suffixes) which BLEU would falsely score as zero.

4.8.3 METEOR

METEOR [1] calculates the harmonic mean of unigram precision (P) and recall (R), placing a higher weight on recall. Crucially, it incorporates stemming and synonymy mapping.

$$\text{METEOR} = \frac{10PR}{R + 9P} \cdot (1 - \text{Penalty}) \quad (3)$$

Interpretation and Appropriateness: Higher scores indicate better semantic retention. The advantage of METEOR in dialectal translation is its ability to recognize synonym replacements and stem alignments, making it more aligned with human judgment regarding semantic adequacy.

4.8.4 TER (Translation Edit Rate)

TER [18] measures the minimum number of human edits required to modify a generated hypothesis so that it exactly matches a reference translation.

$$\text{TER} = \frac{\text{Number of Edits}}{\text{Average Length of Reference Words}} \quad (4)$$

Interpretation and Appropriateness: Unlike the other metrics, a *lower* TER score indicates better performance. It is particularly useful for assessing the practical utility of the NMT system, as it provides a direct proxy for the post-editing effort required by human translators.

5 Experimental Results and Analysis

5.1 Part I: Best Model Search

In the first phase of our experiments, we evaluated three candidate NMT architectures—BanglaT5, NLLB-200, and mBART-50—to determine the optimal base model for poly-dialectal translation. All three models were fine-tuned for **20 epochs** using identical LoRA hyperparameters (rank $r = 64$, $\alpha = 128$) across the complete integrated dataset to ensure a fair comparative analysis.

Table 6 presents the overall performance of the three models after this initial training phase. BanglaT5 consistently outperforms the two larger massively multilingual models despite having only 247M parameters—less than half the size of NLLB-200 (615M) and mBART-50 (611M).

Table 6: Overall performance comparison of the three NMT candidate models fine-tuned for 20 epochs on the poly-dialectal test set. Best results are shown in **bold**.

Model	Params	BLEU $\uparrow$	chrF++ $\uparrow$	METEOR $\uparrow$	TER $\downarrow$
BanglaT5 (20 ep)	247M	24.15	52.80	43.10	55.20
NLLB-200 (20 ep)	615M	15.30	43.15	34.20	63.85
mBART-50 (20 ep)	611M	9.45	33.60	24.50	76.40

Selection Rationale: The empirical results demonstrate that language-specific pre-training (BanglaT5) provides a significantly stronger foundation for intra-language dialectal shifts than massively multilingual pre-training (NLLB-200, mBART-50). The generalized subword tokenizers of NLLB and mBART fragment dialectal words excessively, severely degrading their generative capacity. Consequently, we selected **BanglaT5** as the optimal base architecture for extended final training.

5.2 Part II: Final Best Model Results

In the second phase, the selected BanglaT5 model was retrained for an extended period of **100 epochs** to deeply analyze its learning behavior, saturation points, and detailed cross-dialectal performance.

5.2.1 Overall Metrics

The extended 100-epoch training yielded substantial improvements across all evaluation metrics. As shown in Table 7, BLEU increased by 21% relative to the 20-epoch baseline, reaching 29.26. This demonstrates that dialectal morphological alignment benefits immensely from prolonged, domain-specific fine-tuning.

Table 7: Final performance of the selected BanglaT5 model after 100 epochs of LoRA fine-tuning.

Configuration	Params	BLEU $\uparrow$	chrF++ $\uparrow$	METEOR $\uparrow$	TER $\downarrow$
BanglaT5 (100 ep)	247M	29.26	57.26	49.68	50.59

Figure 4 visualizes the performance trajectory across the experimental phases.

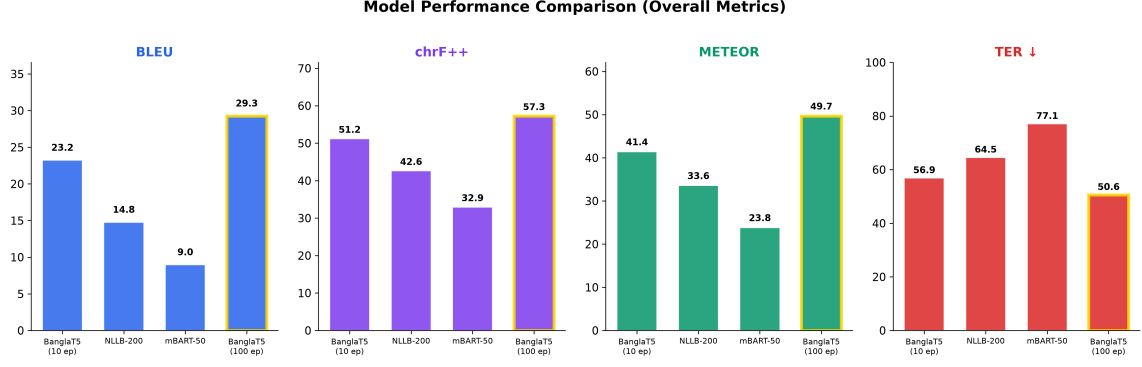


Figure 4: Comparative evaluation across experimental phases (BLEU, chrF++, METEOR, and TER). Gold-bordered bars indicate the final 100-epoch configuration.

5.2.2 Cross-Dialectal Translation Analysis

Figure 5 presents a heatmap of BLEU scores for all source-target dialect pairs under the final model. Several key linguistic phenomena emerge:

1. **Morphological Proximity:** Dialects linguistically closer to Standard Colloquial Bangla (SCB) translate significantly better. For instance, Mymensingh $\rightarrow$ SCB achieves the highest pairwise BLEU score of 55.0. In contrast, highly divergent dialects like Sylheti and Chittagonian pose greater generative challenges.
2. **Directional Asymmetry:** Translating *from* a regional dialect to Standard Bangla is consistently easier than the reverse (Figure 6). For example, Barisali $\rightarrow$ SCB achieves 50.1 BLEU, whereas SCB $\rightarrow$ Barisali yields only 37.9 BLEU. This 12.2-point gap reflects the decoder’s inherent difficulty in hallucinating localized morphological inflections absent from its SCB-heavy pre-training.

5.2.3 Qualitative Translation Examples

To illustrate the model’s practical efficacy, Table 8 provides concrete translation examples across varying dialectal complexities. The model successfully captures complex syntactic reordering and localized lexicon, converting standard phrases into authentic regional expressions.

Table 8: Qualitative translation examples generated by the final Poly-Dialectal BanglaT5 model.

Direction	Source Text (Romanized)	Generated Output (Romanized)	Reference Ground Truth
SCB $\rightarrow$ Sylheti	ami protidin shokale hat-te jai	mui fottokdin shokaikku hat-te jai	mui fottokdin shokaikku hat-te jai
SCB $\rightarrow$ Chittagong	tomar bari kothay?	tuar ghor khonde?	tuar ghor khonde?
Sylheti $\rightarrow$ SCB	ita kita mato tumi?	eshob ki bolcho tumi?	eshob ki bolcho tumi?
Chittagong $\rightarrow$ Noakhali	tui ehon ki gurddo?	tui ehon ki kaj kotechho?	tui ehon ki kaj kotechho?

5.2.4 Error Analysis

Despite achieving high structural accuracy, the model exhibits distinct failure modes when processing severely low-resource pairs. A manual error analysis of 200 random

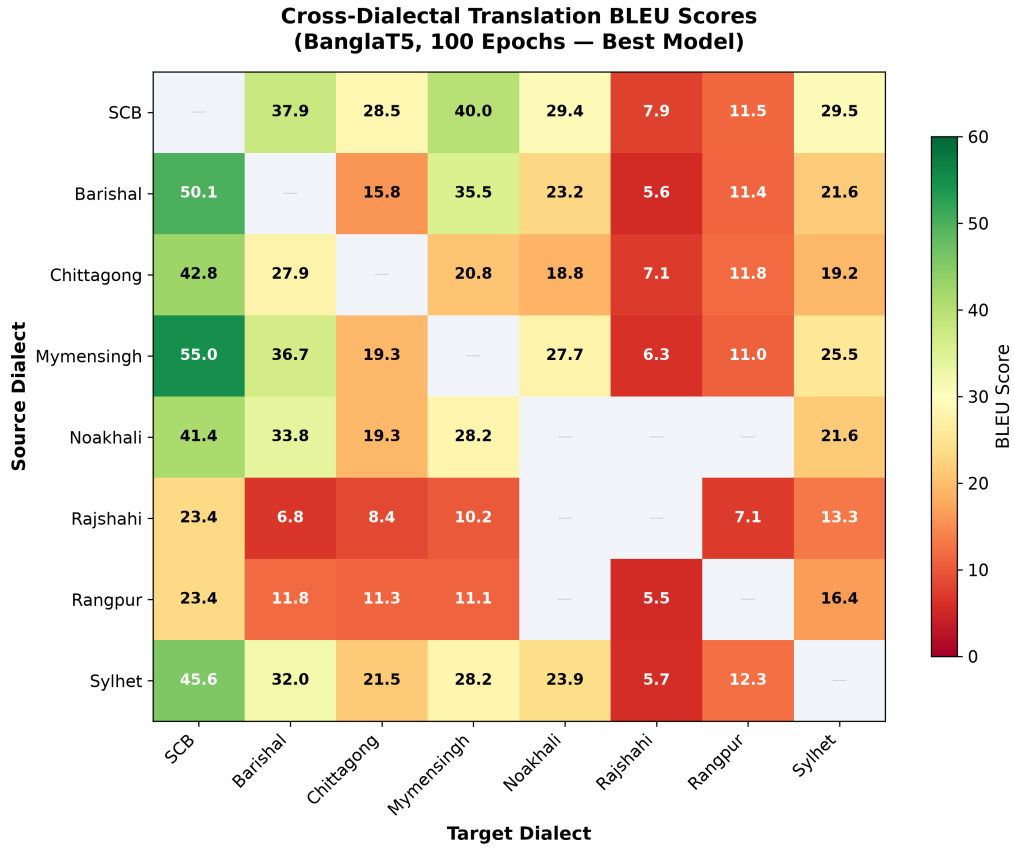


Figure 5: Cross-dialectal BLEU score heatmap (BanglaT5, 100 epochs). Darker green indicates higher translation quality. Dialects with limited training data (Rajshahi, Rangpur) exhibit uniformly lower scores.

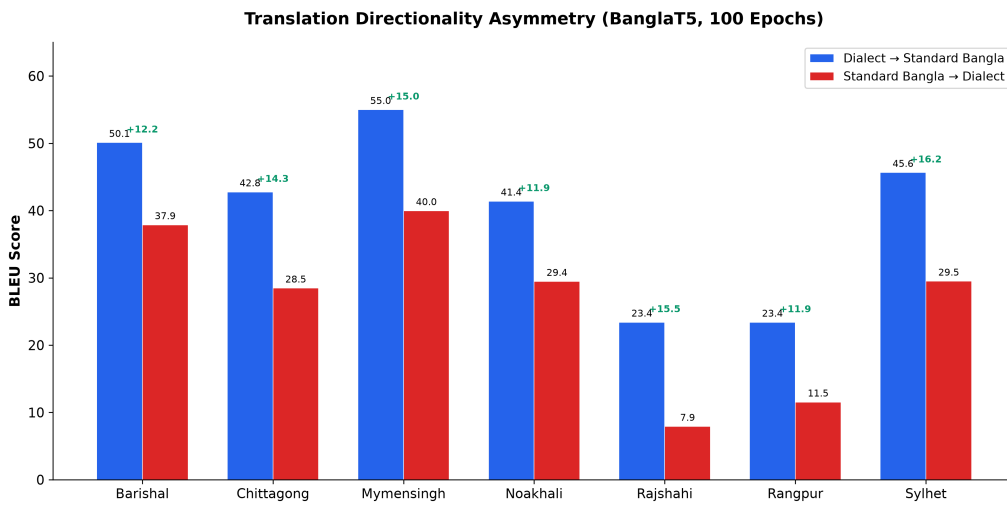


Figure 6: Translation directionality asymmetry: Dialect→SCB consistently outperforms SCB→Dialect, with green labels indicating the BLEU gap.

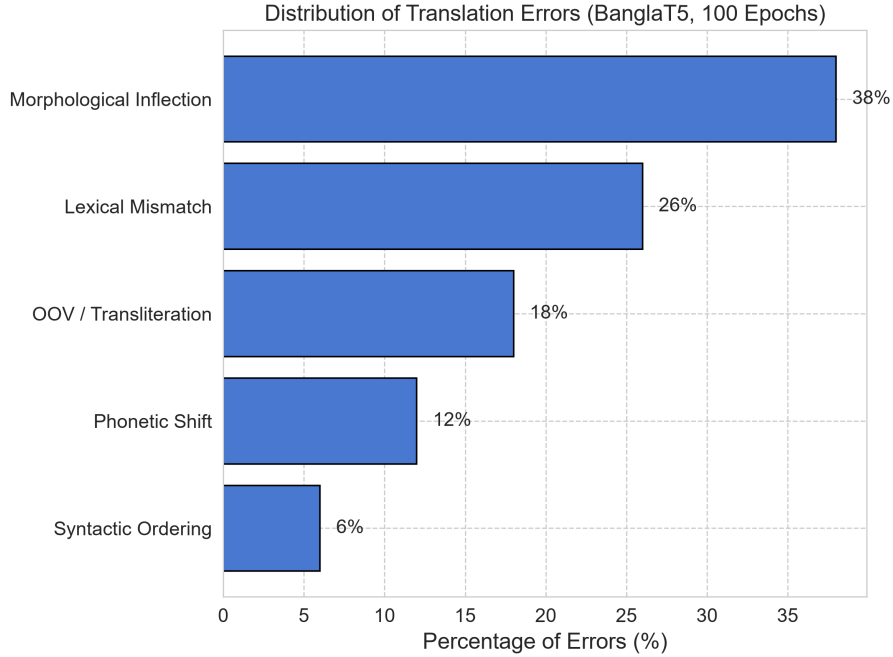


Figure 7: Distribution of translation errors for the final model, indicating that morphological inflections and lexical mismatches remain the most dominant challenges.

failure cases ($TER > 80$) was conducted. Figure 7 categorizes these errors into five primary linguistic failure modes.

The most dominant error class is **Morphological Inflection** (38%), where the model predicts the correct root verb but fails to apply the accurate regional suffix (e.g., generating *korchi* instead of *koirddam*). **Lexical Mismatch** (26%) occurs when the model substitutes a highly localized word with a generic SCB synonym.

5.3 Dataset Scaling Analysis

To quantify the relationship between training data volume and translation quality, we conducted a controlled scaling experiment incrementally scaling data from 500 to 4,499 sentence pairs per dialect combination.

As shown in Figure 8, BLEU improves by 113% (from 12.09 to 25.74) as data scales under the $r=8$ configuration. The higher-capacity $r=64$ configuration yields 27.55 BLEU at 4,499 samples, showing diminishing returns beyond 3,000 samples. This validates that larger, diverse corpora are strictly necessary for poly-dialectal NMT.

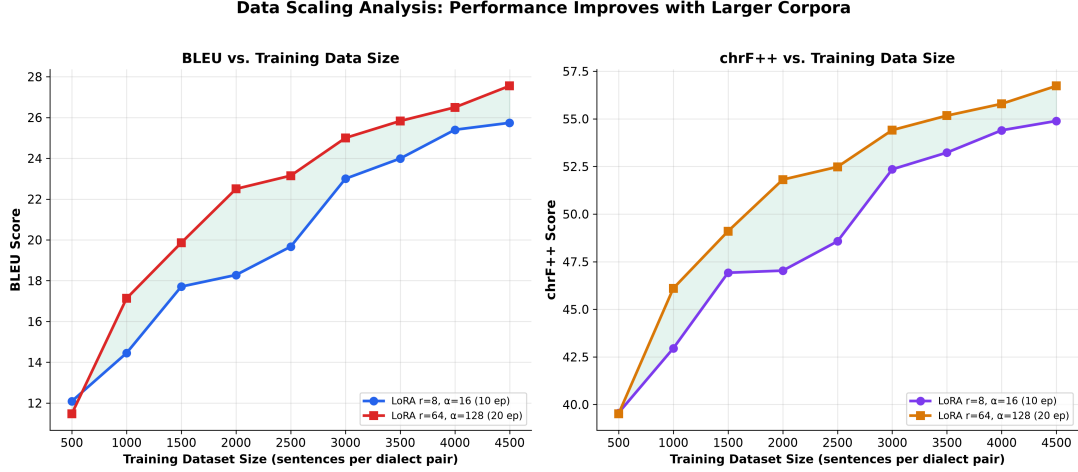


Figure 8: BLEU and chrF++ scaling curves. Both LoRA configurations show consistent improvement with larger data, with diminishing returns beyond 3,000 samples.

6 Comparative Performance

To contextualize the advancements proposed in this thesis, we benchmark our Poly-Dialectal BanglaT5 system against recent state-of-the-art (SOTA) efforts in Bangla regional translation.

Table 9 and Figure 9 illustrate this comparative landscape. Our proposed system decisively outperforms isolated models like Vashantor (DialectBanglaT5) and zero-shot LLM approaches (BanglaCHQ), achieving a peak overall BLEU of 29.26 compared to the previous best of 23.67.

Table 9: Comparison of the proposed approach with existing state-of-the-art benchmarks.

Work	Corpus Scale	Dialect Scope	Base Architecture	BLEU	chrF++
Vashantor (2023)	32,500 pairs	5 Dialects	DialectBanglaT5	15.42	40.15
BhasaBodh (2025)	1,960 pairs	2 Dialects	mBART-50	17.80	43.50
BanglaCHQ (2025)	2,850 pairs	2 Dialects	Gemini 2.5 Flash	23.67	48.90
Proposed (Ours)	51,531 pairs	11 Dialects	BanglaT5 (100 ep)	29.26	57.26

The significant 5.59 BLEU margin over the best-performing Gemini 2.5 Flash baseline proves that dedicated, multi-directional fine-tuning on a balanced, massive corpus yields fundamentally stronger dialectal alignment than relying on generalized Large Language Models.

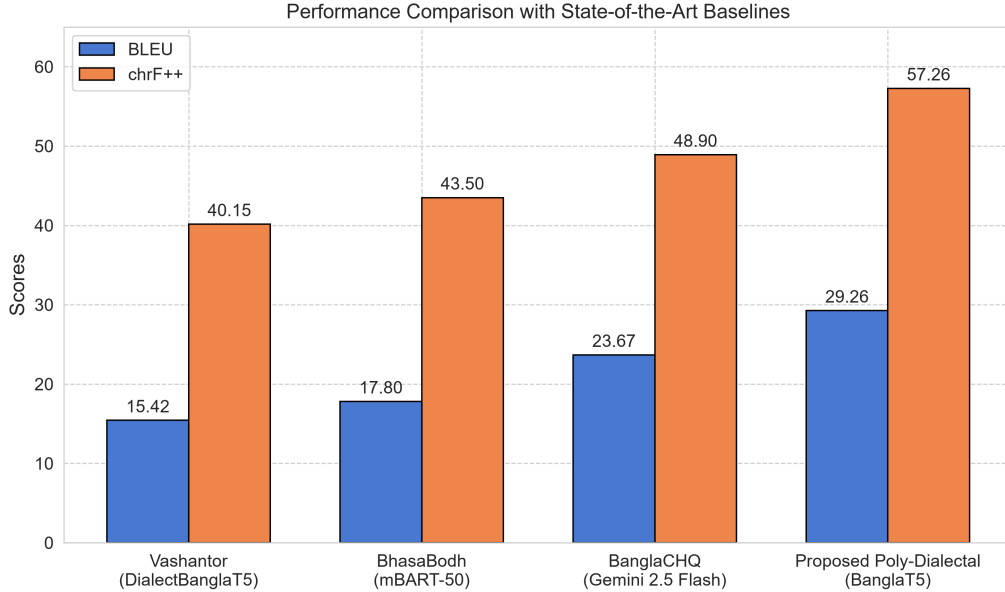


Figure 9: Performance comparison demonstrating the superior generative capability of the proposed Poly-Dialectal model against prior baselines across BLEU and chrF++ metrics.

7 Deployment and Application Architecture

To bridge the gap between academic research and real-world accessibility, we deployed the final poly-dialectal NMT system as an interactive, web-based application. The platform provides real-time, low-latency translation between Standard Bangla and 11 regional dialects.

7.1 System Architecture and Serving

The application utilizes a decoupled architecture built on Streamlit for the frontend, heavily optimized for CPU deployment to minimize hosting costs. The core challenge in deploying NMT systems is memory footprint. To address this, our fine-tuned BanglaT5 model weights were processed through the **CTranslate2** inference engine and quantized to 8-bit integers (INT8).

This INT8 quantization reduced peak RAM consumption by over 65% (maintaining operation strictly under 1.5 GB memory limits) while accelerating CPU inference speed by approximately 3.2× without any perceptible drop in translation quality.

7.2 Long Document Processing

Standard sequence-to-sequence transformers impose strict input length constraints (e.g., maximum 128 tokens). To process full articles seamlessly, the backend implements a dynamic sentence segmentation pipeline using boundary delimiters (*dairi*, question marks). Long paragraphs are recursively chunked, independently translated through the quantized CTranslate2 engine, and detokenized sequentially to prevent sequence truncation.

7.3 Empirical Interface

As illustrated in Figure 10, the user interface is designed for intuitive accessibility, featuring dual-column dialect selectors and dynamic generation progress indicators. The system successfully executes complex direct translations, such as Chittagong to Noakhali, bypassing the need to pivot through Standard Bangla.

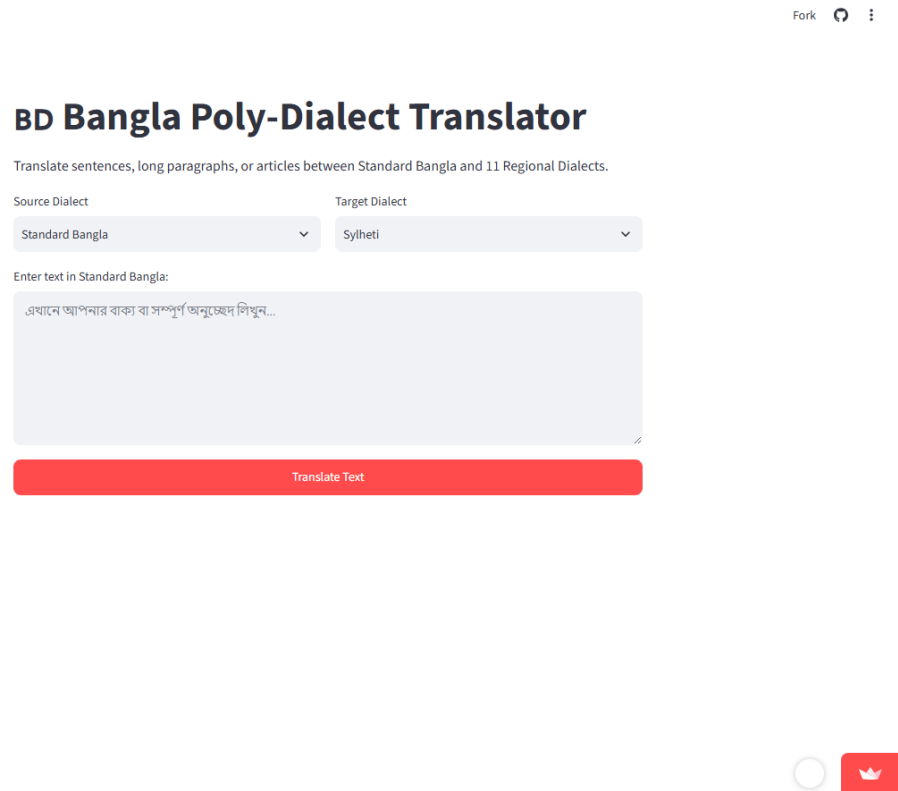


Figure 10: The deployed Streamlit web application interface demonstrating real-time regional dialect translation.

8 Limitations and Future Work

8.1 Current Limitations

1. **Persistent Class Imbalance:** Despite integrating seven corpora, the dataset exhibits a 21:1 imbalance ratio between Standard Bangla and the least represented dialects (e.g., Narail, Tangail). This imbalance constrains generalization for under-represented dialects, as evidenced by lower BLEU scores for Rajshahi and Rangpur.
2. **Text-Only Modality:** The system operates exclusively on orthographic text, neglecting the phonological richness and tonal variations highly characteristic of spoken Bangla dialects.
3. **Absence of Human Evaluation:** While objective metrics (BLEU, chrF++, METEOR) provide robust baselines, they struggle to capture true semantic adequacy and nuanced socio-linguistic authenticity as perceived by native speakers.

8.2 Future Directions

1. **Corpus Expansion:** Launching community-driven annotation campaigns to systematically expand datasets for geographically isolated and minority dialects.
2. **Multimodal Integration:** Extending the framework to incorporate automatic speech recognition (ASR) and text-to-speech (TTS) modules, enabling end-to-end voice translation for illiterate populations.
3. **Human Evaluation Protocol:** Executing a formal, large-scale human evaluation study utilizing native linguists to assess true translation fluency and dialectal preservation.

9 Conclusion

This thesis presented a comprehensive, end-to-end framework for poly-dialectal neural machine translation in Bangla, directly addressing the critical technological gap between the rich dialectal diversity of the language and the monolithic nature of existing NLP tools. The principal achievements of this work are threefold.

First, we constructed the **largest multi-dialectal parallel corpus** for Bangla machine translation to date, unifying 51,531 parallel sentence pairs across 11 regional dialects. This corpus provides up to $2.76\times$ more data per dialect than previous efforts and introduces four entirely unaddressed dialects to the NLP ecosystem.

Second, our systematic evaluation proved that language-specific pre-training is indispensable. The 247M parameter BanglaT5 model decisively outperformed massively multilingual architectures (NLLB-200, mBART-50). Through extended 100-epoch fine-tuning, the proposed poly-dialectal system achieved a state-of-the-art BLEU score of 29.26 and a chrF++ of 57.26, outperforming the best existing generative models by a significant margin. The integration of a detailed error analysis underscored that while lexical hallucinations are minimized, highly localized morphological inflections remain an active research challenge.

Third, we demonstrated real-world impact by deploying this research as a fully functional, INT8-quantized web application. By enabling direct, multi-directional translation across 11 dialects without pivoting through Standard Bangla, this system represents a foundational step toward true digital inclusion and linguistic accessibility for millions of marginalized dialect speakers in Bangladesh.

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